Robert Masen

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Work

Vultr

July 2025 - Current

Staff Software Engineer

- Virtual Machine Metadata Server
  Ported a Virtual Machine Metadata server from PHP to Rust
- Spearhead Rust project workspace organization
  Lead an effort to transition a proof of concept rust hypervisor project to a full cargo workspace
- Ceph guest metrics from host kernel
- Architect Real Time Inventory System
  Designed and developed a system for virtual machine hosts to report guest resource allocation in real time

SmartThings

March 2020 - July 2025

Staff Software Engineer

- Hub Local Metrics
  Implementing a memory efficient, metrics collector for an embedded linux target
- Lua Sandbox
  Develop and maintain 3rd party script sandbox
- Edge HTTP API
  Built an HTTP API running on edge devices to act as a cloud service
- RPC Rewrite
  Swapping out gRPC for a custom RPC solution
- Utility Services
  Develop platform utility services to address shared needs across multiple teams
- Documentation and Design
  Document changes and future opportunities
- Formal Mentor
  Mentored an Associate Software Engineer with the goal of getting promoted

Solution Design Group

February 2019 - March 2020

Software Development Consultant

- Develop ASP.net Application
  Build new features into a ASP.net payment portal
- Modernization and Consistency
  Introduce static analysis utilities to increase code quality and consistency
- Create Community
  Started a Programming Languages Community

United Subcontractors

August 2017 - December 2018

Software Developer

- Develop ASP.net Web Forms
  Build new features and perform bug fixes on an existing ASP.NET Web Forms application
- React
  Transition existing application functionality into a React application using Typescript

Open Source

Rusty ECMA

JavaScript Parsing/Generating libraries

- RESAST
  JavaScript AST in Rust
- RESSA
  JavaScript Parser in Rust
- RESS
  JavaScript Tokenizer in Rust
- RES-Regex
  A parser of JavaScript Regular Expression Literals
- RESW
  JavaScript Writer over RESAST data structures

Cosock

Lua couroutine I/O libraries

- Lustre
  Websockets in Lua